

Resume

Hi, I am Rasmus Kirk. I hold a bachelor's degree in Computer Science from Aarhus University, Denmark, and I am currently pursuing a master's degree with a specialization in Cryptography (specifically SNARKs) and Formal Software Verification, that I will finish March 2026. I have been working part-time as a student worker at Concordium, a cryptocurrency firm, for about three years, primarily writing in Rust.

Here is a list of some of my software interests:

- Decentralized Technologies
- Cryptography
- SNARKs
- Zero-Knowledge Proofs
- Free and Open Software
- Functional Programming
- NixOS
- Linux

Some notable projects:

- Roping in Lasso: An accessible guide to Lasso, which enables lookup arguments from much larger tables than previously possible. Lasso is the primary component of Jolt, the SNARK-based virtual machine (zkVM) that proves correct execution for RISC-V programs via large table lookups
- My Master's thesis: Where I implemented the Halo2 proof system and used it to prove correct blockchain light node catch-up.
- My bachelor's thesis: High Assurance Cryptography - Implementing Bulletproofs in Hacspeg
- A Rust Toy-Blockchain: A toy blockchain project based on the exercises from the course "Distributed Systems and Security" from Aarhus University.
- Halo2 Accumulation in Rust: A Rust implementation of the accumulation scheme used in halo2. The repo also contains a report explaining the background, theory and implementation details of the project.
- Nixarr: A Nixos module that aims to make the installation and management of a home media server as easy, and pain free, as possible.
- Submerger: Combines subtitles from two files into one, with customizable position and color settings.

- Rustle: An open source rust program that keeps your speakers active.

Feel free to explore my projects and reach out if you have any questions.

Contact

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